with different algorithms for transitions

Operation	Mean Cost (α=0.98, T_0=200 and k=1000)
Adjacent Swaps	1937.291
Random Swaps	1683.563
Random Permutation	1831.886

It’s interesting that the best result is obtained with the medium-range search, while the worst one is given by the local search. This means that, with the configuration used, the local search gets stuck in local optima, while the random permutation is too erratic and fails to obtain local convergence in the final stage of the algorithm, when temperature gets low.

We must stress that, with different parameters, this could change completely. Take, for instance, α, the decaying factor: with a difference choice for the cooling process would be slower, so we wonder, could this allow the local search to work better? Let’s try it out – results in table 17.2.

Table 17.2 Average cost, with different temperature decay rate

Operation	Mean Cost (α=0.97)	Mean Cost (α=0.98)	Mean Cost (α=0.99)
	1972.502	1937.291	1868.701